Made in Katana: Creating immersive digital music experiences with zero downtime on Google Cloud

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About Made in Katana

Made in Katana a
record 3 million+
with zero crashes
creative team 10
DevOps tasks, an
innovative fan en
features with App
and Firestore.

About Made in Katana

Made in Katana is a global creative and digital marketing agency offering unique insight into the entertainment and youth industries. MIK's groundbreaking creative solutions building websites, podcasts, videos, graphics, and social campaigns have attracted major players such as Spotify, Warner Music, and Paramount Pictures as clients. Since 2015, MIK has produced Australia's The Hottest 100, an iconic online music poll, taking fan engagement to record levels almost every year.

Google Cloud results

- Promotes continuous creative platform innovation by reducing DevOps from 30% of engineering time to zero
- Deepens fan loyalty with 100% platform reliability due to unlimited scalability of App Engine and Firestore database
- Enables global music audience reach by spinning up 100 server instances in less than 10

Zero p
down
record
traffic

Industries: Media & Entertainment

Location: Australia

Google Cloud (https://cloud.google.com/)

help.

App Engine (https://cloud.google.com/appengine/)

Cloud Firestore (https://cloud.google.com/firestore/)

seconds on App Engine

- Inspires new user engagement strategies with world-class AI solutions and proactive collaboration

Every January, Australians take a day off from their everyday lives to tune into The Hottest 100 (https://www.abc.net.au/triplej/hottest100/20/). Part music countdown, part national ritual, people spend the day listening to the results of an online poll in which millions have voted, waiting to hear where their favorite song of the past year ranks.

Since digital creative agency Made in Katana (https://madeinkatana.com/) (MIK) took charge of producing the event in 2015, Hottest 100 keeps breaking its voting record. That's thanks to a creative team that ramps up user experience with interactive features and cutting-edge design. The number of votes for the 2020 countdown topped three million (https://www.abc.net.au/triplej/news/musicnews/the-2019-hottest-100-breaks-voting-record-three-million/11889756) for the first time, for a 16% jump from the previous year.

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> *—Adam Callen, co-founder and Creative Director, Made in Katana*

The excitement of seeing an already massive event go off the charts can also be nerve-racking. Millions of fans flooding the system can severely strain databases and computing resources, causing crashes that frustrate users. Any glitch will turn off vast swaths of the public, throwing cold water on the event.

To produce an experience that literally grows at planet scale (Hottest 100 spreads beyond Australia to as far as Japan and Italy), MIK needed a cloud database that didn't require round-the-clock monitoring to prevent crashes. It also sought scalable computing power that allowed limitless server requests, not only for Hottest 100 but also projects such as artist content for music streaming service Spotify (https://www.spotify.com/us/). The websites MIK makes for Spotify can soar to millions of hits per day.

MIK turned to Google Cloud (https://cloud.google.com/) and found the versatile, scalable solutions it needed in a combination of Firestore (/firestore) and App Engine (/appengine). It won peace of mind that no matter what the demands of global pop music fandom, its systems can absorb the tide and keep music lovers happy and engaged.

"After we moved to Firestore for our Hottest 100 database and App Engine for platform development, we went from panic over fan voting spikes to not worrying about a thing, since Google scales automatically. We were also freed to focus on better fan experience," says Adam Callen, co-founder and Creative Director at MIK.

"Thanks to Firestore and App Engine, our wider team, including designers and marketers, suddenly felt safer going to market with new initiatives to drum up the fan base, because we knew the system would be able to handle any load. Google Cloud had positive knock-on effects for our creativity."

—Adam Callen, co-founder and Creative Director, Made in Katana

Cultivating viral music fandom growth with Firestore

When MIK was tapped to run Hottest 100 by radio station Triple J (https://www.abc.net.au/triplej/), the event was still hosted on the station's servers. At that time, organizers sometimes needed to stop accepting votes for a few days, so they could build more capacity to handle higher fan volumes.

One of MIK's first changes was migrating Hottest 100 to a public cloud platform. That initially solved the traffic problems. But Callen and his team soon hit a new problem. The provider's traditional relational database solution was too rigid to handle the explosive audience momentum unleashed by the success of their first Hottest 100 production.

The primary issue was that the provider's SQL (structured query language) database solution scaled vertically, on a single server. That limited the amount of writes, or data inputs, it could handle without being reconfigured.

"We designed our first Hottest 100 with a bigger audience in mind, but that first year still far exceeded our expectations," says Callen. "It meant with our old cloud provider we were already hitting limitations on our servers for the voting database. We realized we needed to change fast, or we'd suffer crashes the next year."

That's when MIK discovered Google Cloud and opened a world of unlimited digital entertainment potential. Through proactive support from the

Google Cloud team, Callen's development team learned how Firestore's non-relational database solution, Firestore in Datastore mode (/datastore/docs), combined with App Engine, could instantly transform scaling capabilities. That led to a full-stack migration to Google Cloud.

The power of Firestore is that its non-relational database (also known as a NoSQL database) is designed specifically for huge scale on a distributed computing network. Firestore scales horizontally across multiple servers, each sharing part of the load. Essentially, the NoSQL database allows Hottest 100 fandom to grow at unlimited pace and volume with the system never crashing.

After migrating to Google Cloud, Hottest 100 experienced zero system downtime and was confident that they would not face any glitches it saw in previous editions, in a year that saw voting soar to record numbers. "It really led to quite spectacular gains for us," says Callen.

"With Google Cloud, the experience for the end user is less frustrating and a lot more seamless and immediate, resulting in fewer users abandoning the voting process. It's a key strategic success we've gained through a combination of Firestore and App Engine."

—Luke Larsen, Senior Web Developer, Made in Katana

Delighting music fans by freeing up creative developers to improve user experience

The ultimate benefit that MIK won from Google Cloud adoption was freedom and confidence to aim for the stars. Suddenly the entire team was liberated to focus on making the platform as outstanding as it could possibly be, without worrying about DevOps.

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That in turn led to a better user experience and more fan loyalty. "With Google Cloud, the experience for the end user is less frustrating and a lot more seamless and immediate, resulting in fewer users abandoning the voting process," says Luke Larsen, Senior Web Developer at MIK. "It's a key strategic success we've gained through a combination of Firestore and App Engine."

One specific example of the newfound ability to delight music fans was restoring Hottest 100's popular "shortlist" feature, allowing fans to select and constantly update their five favorite songs, which MIK introduced with its original provider but had to abandon when it threatened to crash the system.

Imagine music lovers across Australia continuously adding, deleting, bumping up, and bumping down songs on their favorite song shortlist. Each action is a server call. Multiplied exponentially, the strain can build relentlessly as the competition heats up. That is, without a modern NoSQL database.

When MIK switched to Firestore, it decided to resume the shortlist function and see what happened. It was delighted to find that even with much higher fan numbers, the non-relational database handled the load without skipping a beat.

"With the Hottest 100 shortlist, fans are free to essentially spam the database with requests," says Matthew Prasinov, Full Stack Developer at Made in Katana. "Making the transition to Firestore allowed users to do that without us having any concerns of the database breaking down."

Better developer experience with App Engine's power and ease of use

An intuitive, user-friendly interface was another reason why MIK switched to Google Cloud. Previously, the MIK development team spent much of its day configuring settings and deploying multiple unwieldy products for a single operational task.

By switching to App Engine, Larsen says, MIK discovered streamlined development and deployment protocols that saved the team a huge amount of time and headache. The engineering team has gone from spending 30% of its time on

DevOps tasks with its previous provider to nearly zero. That translates into close to 100% of engineering time spent on product improvement.

Moreover, App Engine provides instant unlimited computing power, essential for a platform that reaches millions of music fans around the world. Prasinov says App Engine regularly spins up 100 new instances or more in less than 10 seconds, absorbing any traffic volume.

Now MIK looks forward to an unlimited future of breakthrough digital content through AI tools such as Vertex AI (/vertex-ai) and BigQuery ML (/bigquery-ml/docs), which will enable MIK to leverage its massive data pool to create more unforgettable fan experiences.

"The 'Katana' in our name means sword in Japanese, and that relates to impact that's meant to last and not just be forgotten," says Callen. "We're excited to be working in creative ways with the supportive Google team to forge an enduring edge in digital content creation with Google AI tools."